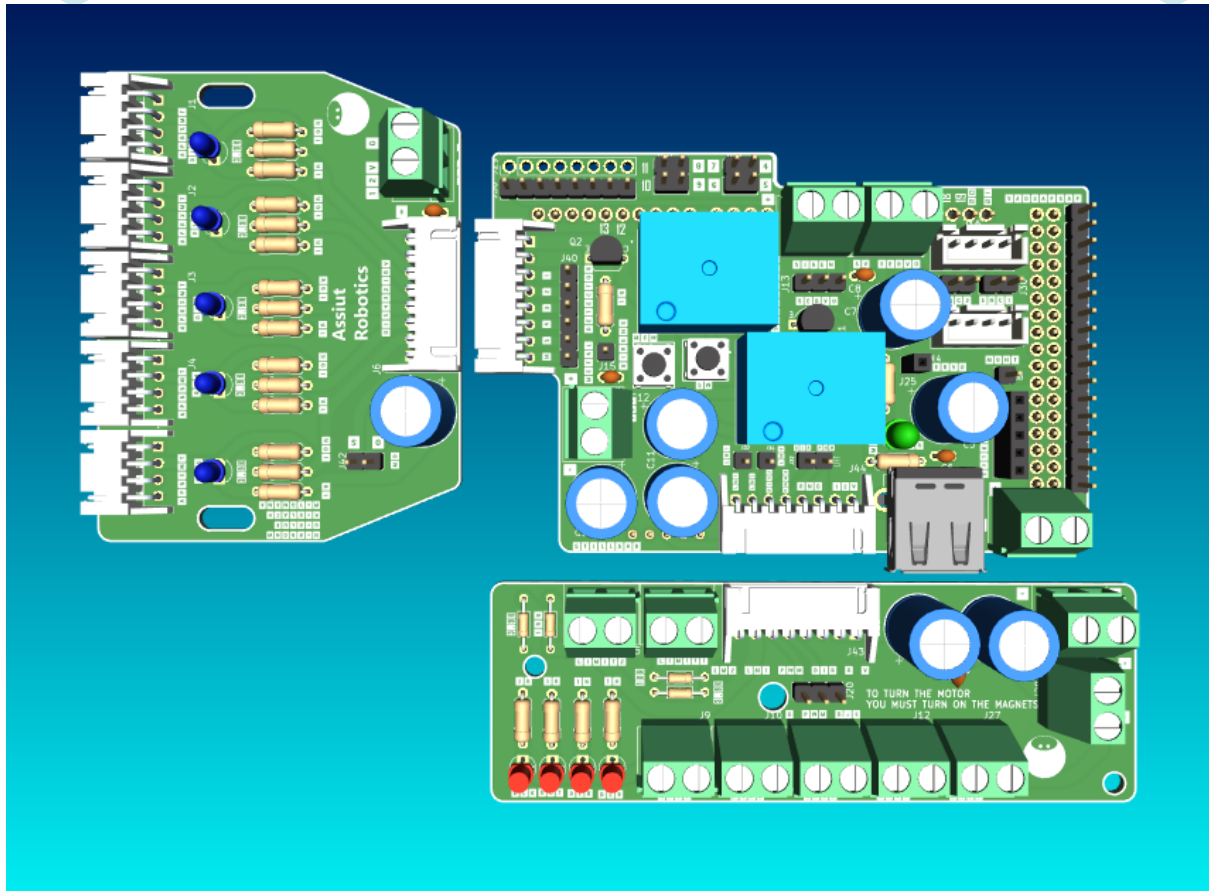


SYSTEM SCHEMATIC & PCB



Schematic Overview:

System is divided into 3 sub-modules to reduce long wire and each module connected to the other by S8B JST-Connector.

Metal Detector:

The first module is the Metal detector module which contains the metal detector interface + 5 Proximity sensors and LED for indication.

Each proximity sensor has **4 wires** interface and with the **2 power pins** and **2 pins** for metal detector signals the total of wires entering this board is **24 wires** and then reduced only to **8** by the **JST connector**.

Lift Mechanism:

The mechanism used to gather the mines contains magnets and a DC motor to move an arm with limit switches for control.

This module has **10 wires** for magnets (2 each) and **2 for the DC motor** and **2 for the DC motor control** + **2 limit switches**.

Total: **16** and then ultimately reduced to **8** by the **JST connector**.

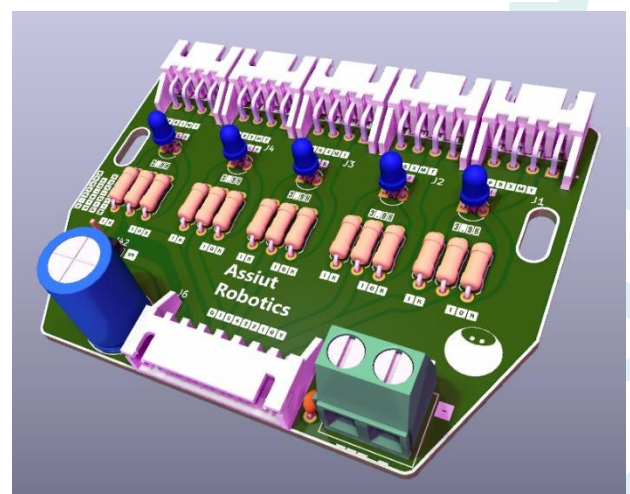
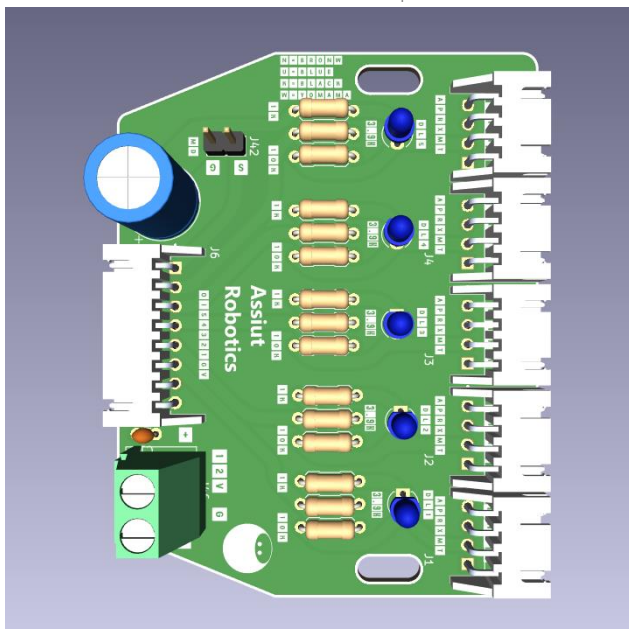
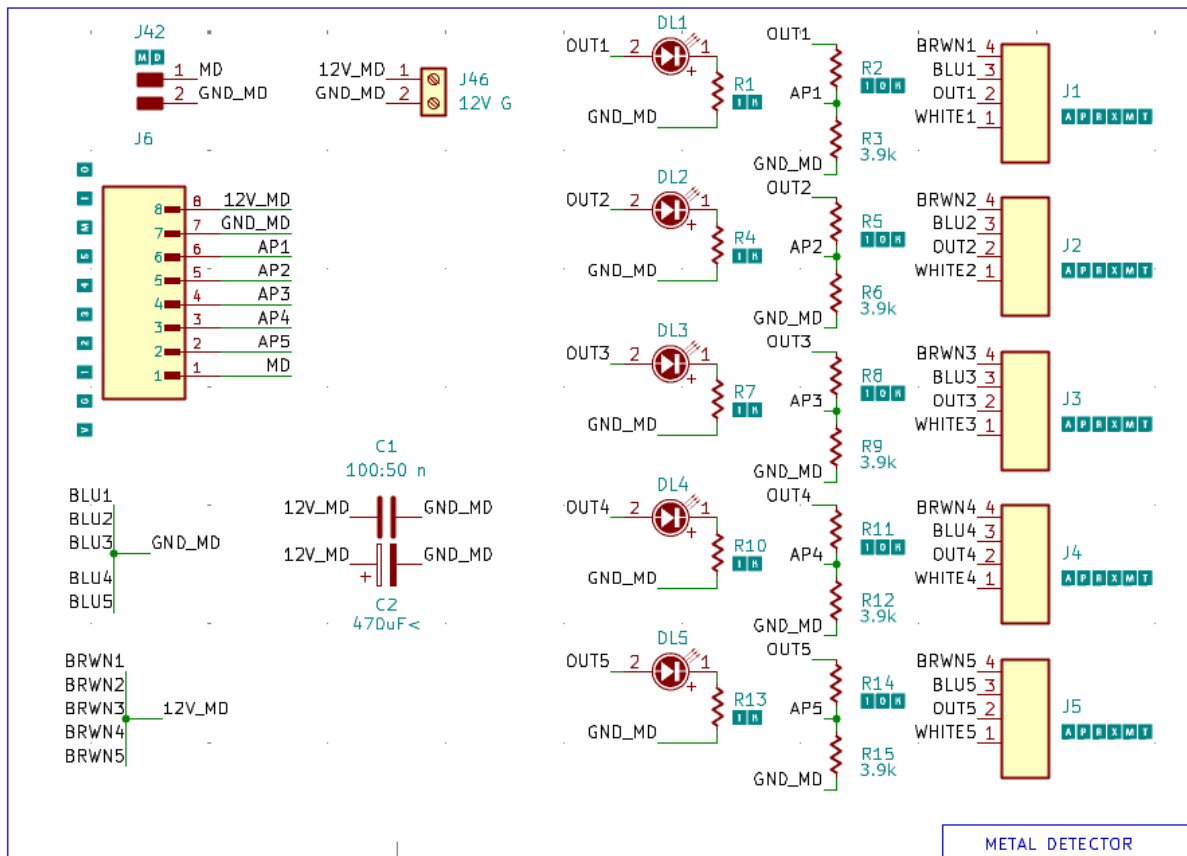


Centralized Module:

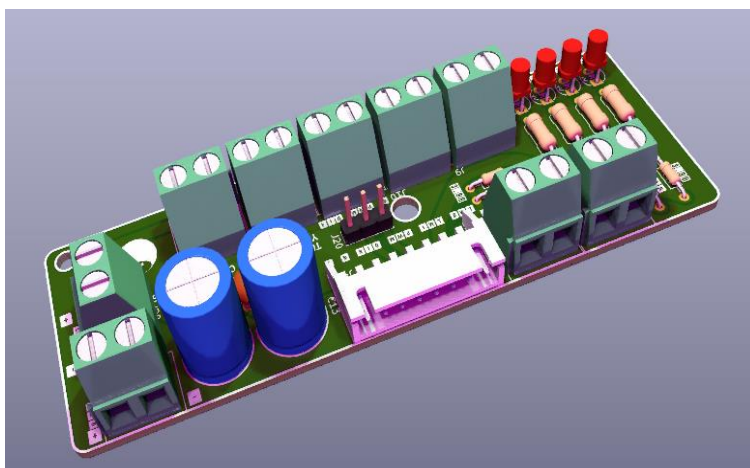
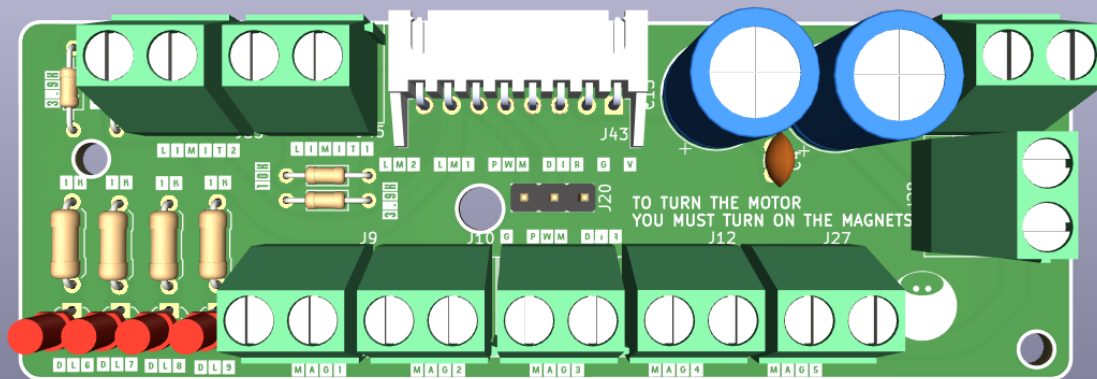
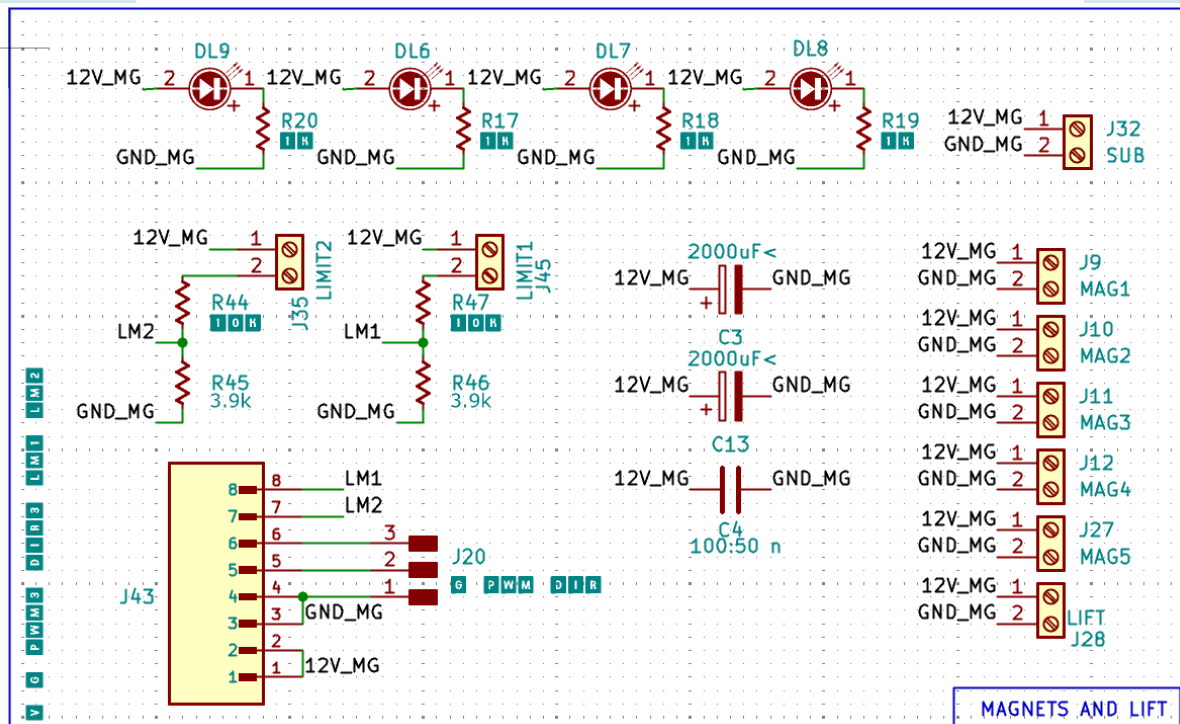
This module is attached directly on the **Arduino MEGA** which is the main low-level controller so the module is shaped like a shield attached directly on top of it and has the main power handling and the **Raspberry PI** interface.

The module also has a socket for each sensor pin to allow external interface if any other microcontroller is chosen except the Arduino which enables flexibility.

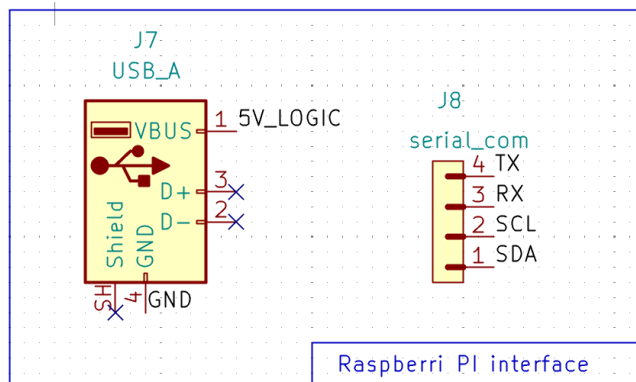
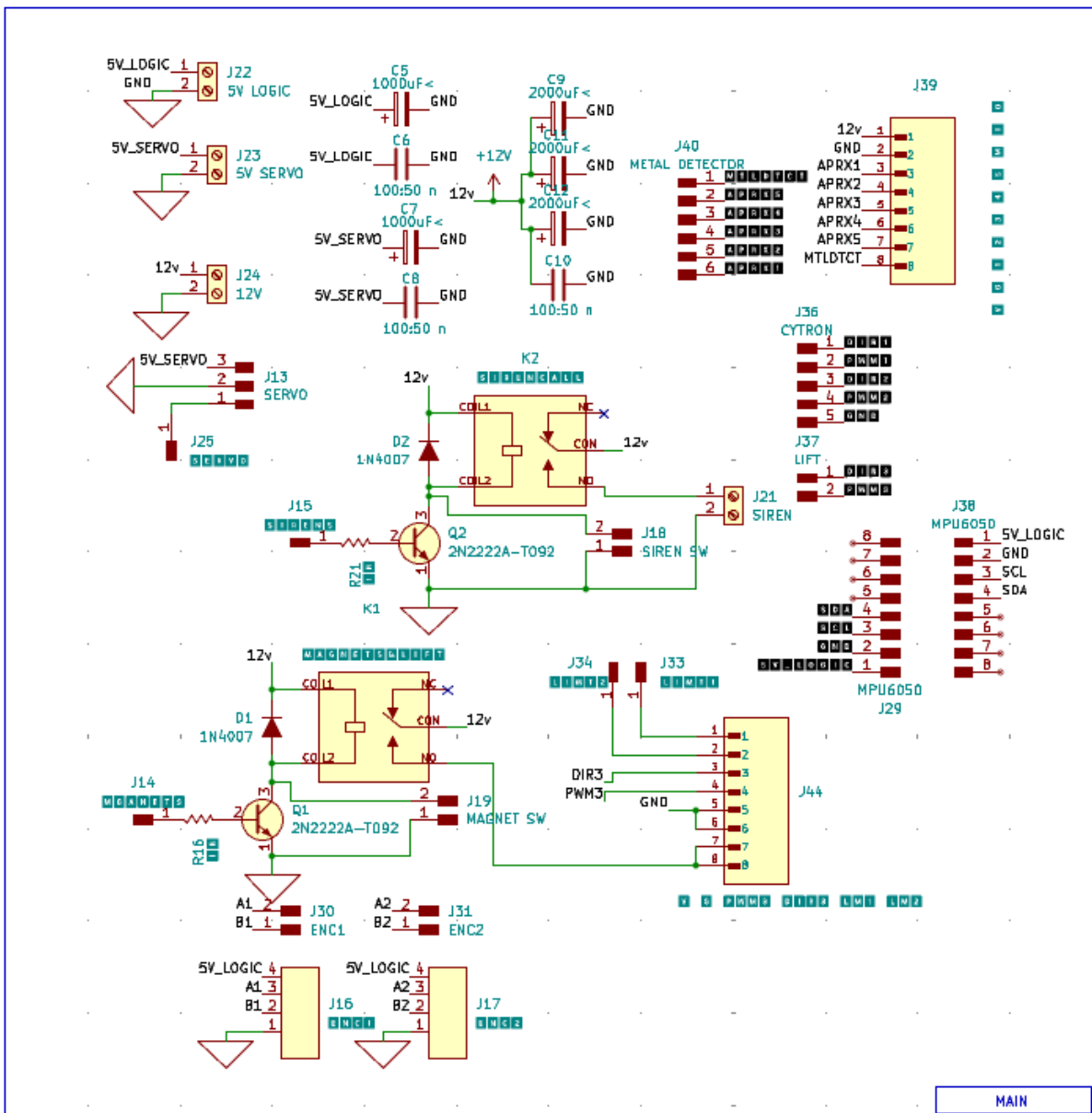
Schematics and Renders:



METAL DETECTOR

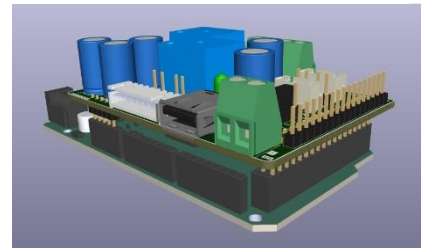
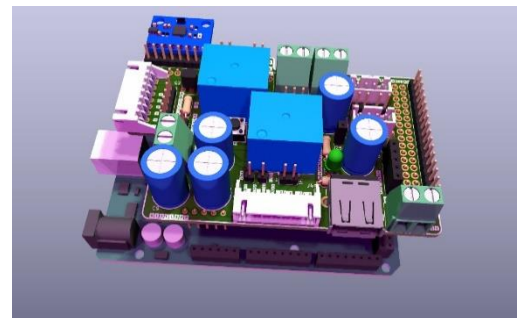
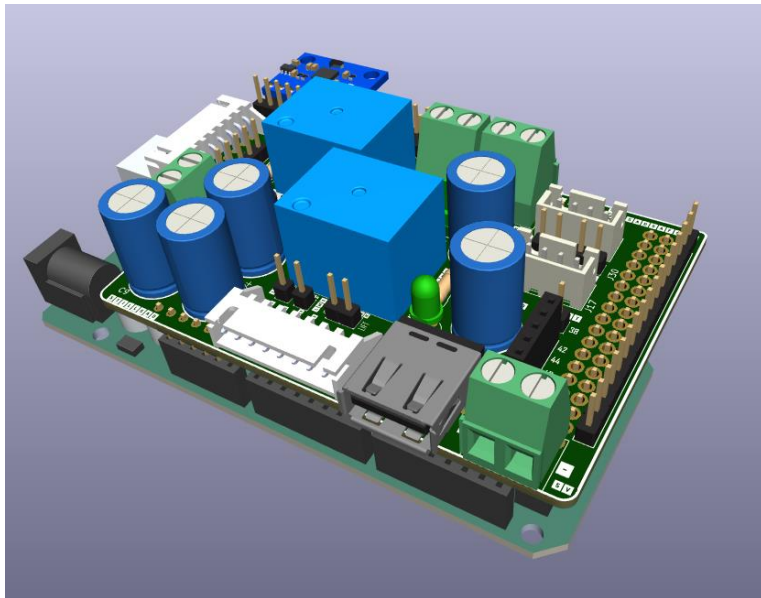
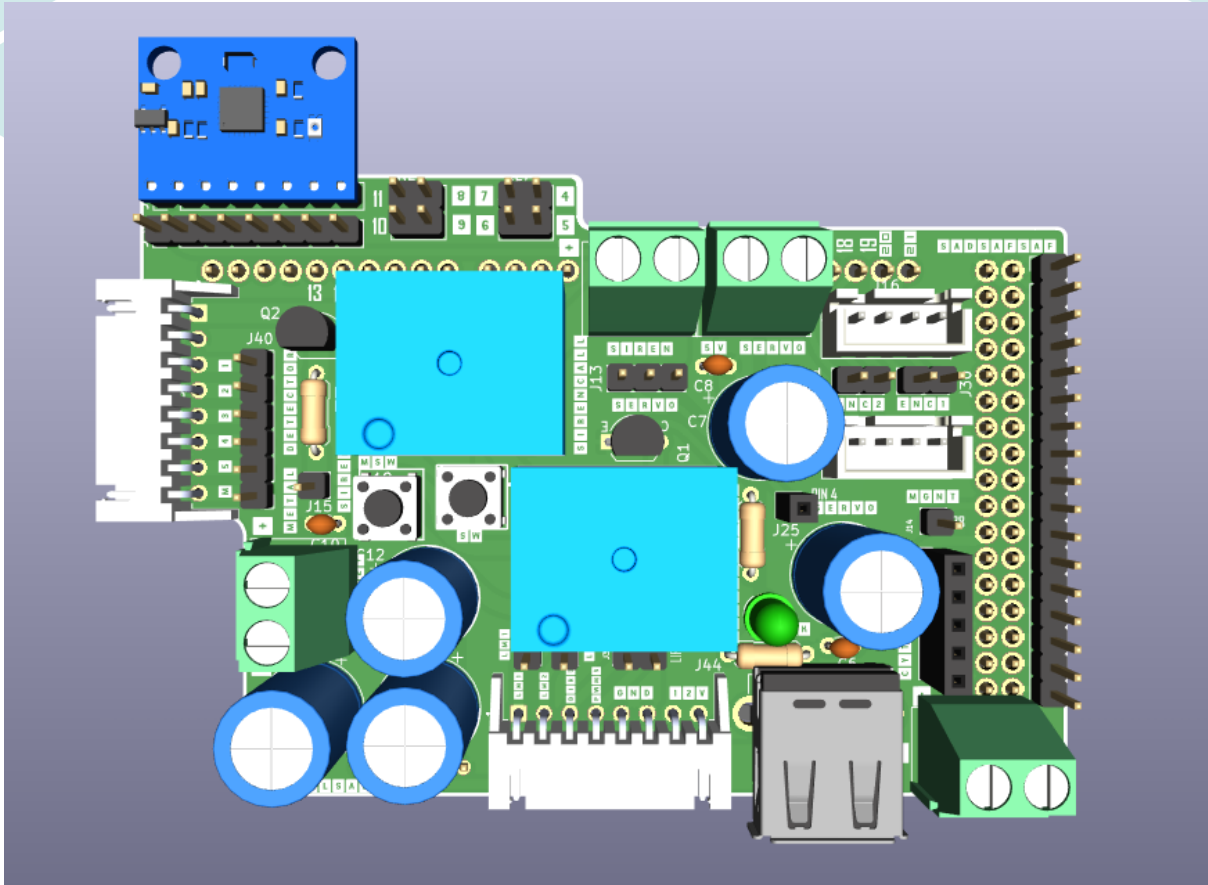


LIFT MECHANISM





MINESWEEPERS
TOWARDS A LANDMINE-FREE WORLD



**CENTRAL
MODULE**

